

SET perihelion derivation 13

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We derive the spherical exterior and its perihelion advance from SET's capacity relation and source normalization within a specified local metric completion. Common matter and field evolution restricts an initially arbitrary momentum dependence to the quadratic kinetic form. Requiring that spherical response to come from a smooth three-dimensional metric response then selects its remaining radial function. The exterior field follows from the action, and every exterior screen reads the same source charge as a consequence.

The derivation proceeds from the invariant through action selection, conserved energy, normalized readout and orbital motion to the Mercury benchmark. The local kinetic coefficients use only the physical spatial metric, with the spatial derivative class and common evolution requirement stated below. These conditions select the spherical Einstein action; the scalar capacity relation alone does not classify all field theories.

We consider an isolated, nonrotating spherical source and a test body in its vacuum exterior, with $r > R_s = 2GM/c^2$. Source multipoles, other bodies and radiation reaction are outside this orbit calculation.

1 The capacity invariant and radial spatial measure

Start from the SET invariant [10] and define the event-progression ratio:

$$S^2 + V_{\text{rel}}^2 + c^2 \beta^2 = c^2, \beta = \frac{d\tau}{dT}. \quad (1)$$

Here T is the stationary reference time normalized at infinity, τ is the test body's recorded time, and S is gravitational burden in speed units. Let r be areal radius, defined by the area $4\pi r^2$. Spherical symmetry permits an initially unspecified positive radial coefficient q :

$$V_{\text{rel}}^2 = q(r) \left(\frac{dr}{dT} \right)^2 + r^2 \left(\frac{d\Omega}{dT} \right)^2, q > 0. \quad (2)$$

The angular abbreviation is $d\Omega^2 = d\theta^2 + \sin^2\theta d\varphi^2$. Define α and h without yet fixing the spatial response:

$$\alpha^2 = 1 - \frac{S^2}{c^2}, h = \alpha \sqrt{q}, q = \frac{h^2}{\alpha^2}. \quad (3)$$

Multiplying the invariant by dT^2 gives the stationary interval:

$$c^2 d\tau^2 = \alpha^2 c^2 dT^2 - q dr^2 - r^2 d\Omega^2. \quad (4)$$

A held clock records $d\tau_{\text{hold}} = \alpha dT$ and radial rulers measure $d\ell = \sqrt{q} |dr|$. We require $\alpha \rightarrow 1$ and $q \rightarrow 1$ at infinity. The field variation below determines their radial dependence.

2 Source normalization and the dynamical variables

For a specified source domain Ω , retain the source law and its radius-independent gravitational normalization:

$$Q_\Omega^2 = 24 \pi G M V_\Omega, \mu \equiv \frac{Q_\Omega^2}{24 \pi V_\Omega} = G M. \quad (5)$$

For a spherical reference domain of radius R_0 , substitute $V_\Omega = 4 \pi R_0^3 / 3$:

$$Q_\Omega^2 = 32 \pi^2 \mu R_0^3, Q_\Omega = 4 \pi \sqrt{2 \mu R_0^3}. \quad (6)$$

The energy identification used in this completion is

$$E_Q = \frac{c^2 Q_\Omega^2}{24 \pi G V_\Omega} = M c^2. \quad (7)$$

Here M denotes total gravitational mass, including internal and binding-energy accounting. It need not equal the sum of the source's constituent rest masses. Sections 7–10 connect this source energy to the Hamiltonian energy of the selected action. Changing a reference radius changes its corresponding Q ; the conserved normalization is Q^2 / V at fixed M .

To test the action before imposing stationarity, introduce a time coordinate t and radial label x . The areal radius $R(t, x)$ may change. The general spherical interval takes the form

$$ds^2 = N^2 c^2 dt^2 - \Lambda^2 (dx + u dt)^2 - R^2 d\Omega^2. \quad (8)$$

The lapse N relates normal clock progression to dt ; the shift u describes radial relabeling between slices. On a t -constant slice, Λdx is radial distance. A dot and prime in the field calculation denote ∂_t and ∂_x . The combinations

$$U_R = \dot{R} - u R', U_\Lambda = \dot{\Lambda} - (u \Lambda)' \quad (9)$$

remove the change due to that relabeling. We keep N and u in the action during variation; they impose the energy and radial momentum constraints. In the static areal chart, $t = T$, $x = R = r$, $u = 0$, $N = \alpha$ and $\Lambda = \sqrt{q}$. The general lapse N is not automatically the held-clock factor α in a moving chart.

The complete isolated system has a regular center and one asymptotically flat end, with $N \rightarrow 1$ and $u \rightarrow 0$. These conditions fix the reference clock and energy boundary term. The exterior solution can also describe the outside of a collapsed source; this note does not vary an excised inner horizon boundary.

3 Capacity universality and common evolution

Capacity universality can become a restriction on how the field is allowed to evolve. We give that idea a dynamical meaning in the spherical construction.

Changes of the field are physical events. Their evolution must use the same operational clocks and distances as material events. Different descriptions of the same physical history must agree about what happens.

First identify what the invariant already supplies. In a stationary region with $\alpha > 0$, a held observer's clock and local distance measurement define

$$d\tau_{\text{hold}} = \alpha dT, v_{\text{local}} = \frac{d\ell}{d\tau_{\text{hold}}}, b_{\text{local}} = \frac{d\tau}{d\tau_{\text{hold}}}. \quad (10)$$

The invariant uses $V_{\text{rel}} = d\ell / dT$ and $\beta = d\tau / dT$. Express both through the local readings:

$$V_{\text{rel}} = \alpha v_{\text{local}}, \beta = \alpha b_{\text{local}}. \quad (11)$$

Substitute into the capacity relation and use $c^2 - S^2 = c^2 \alpha^2$:

$$\alpha^2 (v_{\text{local}}^2 + c^2 b_{\text{local}}^2) = c^2 \alpha^2, \quad (12)$$

$$v_{\text{local}}^2 + c^2 \left(\frac{d\tau}{d\tau_{\text{hold}}} \right)^2 = c^2. \quad (13)$$

Given the physical distance norm, this is the common local clock-and-motion relation. It does not require $q = \alpha^{-2}$; the radial response remains to be determined by the field equations. It also does not identify a general coordinate lapse with the stationary held-clock factor.

To extend this relation to field evolution, we require the same operational geometry for matter and for changes of the field. Local changes of time slicing describe the same history, with no independent physical clock or foliation field. In the canonical calculation this means that evolution generators must represent the corresponding geometric changes consistently [8,12].

This requirement is stronger than sharing clock units or respecting a maximum speed. Those conditions alone do not specify a propagator or an action. Here the requirement has a definite test: composing two local normal advances in opposite orders must give the spatial relabeling fixed by the common distance measure. The next section states that test; Appendix B derives its geometric coefficient.

4 The field action before kinetic selection

Use $G_* = c = 1$ in the field calculation through Section 8 until SI units are restored. Here G_* labels the action's overall coupling; Section 8 relates it to the measured G in the source law. The canonical pairs are (Λ, P_Λ) and (R, P_R) . Write $p = P_\Lambda$, $q_m = P_R$ and $y = q_m / \Lambda$. The momentum q_m is unrelated to the radial metric coefficient q . The canonical action is

$$I = \int dt dx [p \dot{\Lambda} + q_m \dot{R} + L_{m, \text{canonical}} - N H - u D] + I_\partial. \quad (14)$$

Here $H = H_g + H_m$ and $D = D_g + D_m$. Under radial relabeling, R, p, y are scalars and Λ is a density. Allow an arbitrary smooth local momentum function:

$$H_g = \Lambda k(R, p, y) + V_g, D_g = q_m R' - \Lambda p'. \quad (15)$$

No polynomial degree or time-reversal parity is imposed on k . Retain the spatial derivative form

$$V_g = \frac{F(R) R''}{\Lambda} - \frac{F(R) R' \Lambda'}{\Lambda^2} + \frac{Z(R) (R')^2}{\Lambda} + W(R) \Lambda. \quad (16)$$

The coefficient F has length units and is nonzero on the domain; Z, W, k are dimensionless. The areal radius remains physical. This class excludes momentum derivatives, mixed momentum-gradient terms, lapse derivatives, additional fields and more general spatial dependence [11]. The spherical kinetic law must also admit a smooth local three-dimensional completion whose quadratic coefficients use only the physical spatial metric. Section 8 applies this requirement after the kinetic form has been derived.

A minimally coupled massless scalar tests the common matter geometry [2]. Absorb its angular normalization into ψ and Π ; it is a matter field, not a gravitational radiation mode:

$$H_m = \frac{\Pi^2 / R^2 + R^2 (\psi')^2}{2 \Lambda}, D_m = \Pi \psi'. \quad (17)$$

Define the normal and spatial generators and their lapse combination:

$$H[N] = \int N H dx, D[v] = \int v D dx, \omega = N_1 N_2' - N_2 N_1'. \quad (18)$$

For radial distance Λdx , common geometric evolution requires

$$\{H[N_1], H[N_2]\} = D[\omega / \Lambda^2]. \quad (19)$$

This identity is imposed on arbitrary canonical configurations before the constraints are solved. It is the strong representation requirement, which is more specific than agreement only on physical solutions [12]. Spatial relabeling brackets follow from the density structure above.

5 Common evolution determines the kinetic form

Appendix B evaluates the full bracket with arbitrary k . The gravitational coefficient of ω is

$$B_g = \frac{1}{\Lambda} \left[F k_{yp} p' + F k_{yy} y' + R' A \right], \quad (20)$$

$$A = -F k_p + (2Z - F_R) k_y + F k_{yR}. \quad (21)$$

Subscripts denote partial derivatives, and $F_R = dF/dR$. Matter contributes D_m/Λ^2 with unit coefficient. The required gravitational part is

$$\frac{D_g}{\Lambda^2} = \frac{y R' - p'}{\Lambda}. \quad (22)$$

The derivatives p' , y' and R' are independent in this comparison. Matching their coefficients gives

$$k_{yy} = 0, k_{yp} = -\frac{1}{F}, \quad (23)$$

$$-F k_p + (2Z - F_R) k_y + F k_{yR} = y. \quad (24)$$

The first equation makes k linear in y . Write $k = A(R, p)y + B(R, p)$. The second equation integrates with respect to p to give

$$A(R, p) = -\frac{p}{F(R)} + C(R). \quad (25)$$

Insert this expression in the third equation. The terms proportional to y cancel, leaving

$$B_p = \frac{2(F_R - Z)}{F^2} p + C_R + \frac{2Z - F_R}{F} C. \quad (26)$$

Integrating once more gives the general local solution:

$$k = -\frac{py}{F} + \frac{F_R - Z}{F^2} p^2 + Cy + E_C p + U(R), \quad (27)$$

$$E_C = C_R + \frac{2Z - F_R}{F} C. \quad (28)$$

The functions C and U are arbitrary at this stage. The required geometry has forced at most quadratic momentum dependence, although the initial function allowed arbitrary powers and nonpolynomial terms. The remaining linear terms must be examined before identifying a distinct physical response.

Equivalent momentum descriptions

The linear terms in the preceding solution can be removed by a canonical change of momentum origin. Define

$$\sigma(R) = F(R)C(R), J[\Lambda, R] = \int \Lambda \sigma(R) dx. \quad (29)$$

The functional derivatives of J give the new momenta:

$$\bar{p} = p - \sigma, \bar{q}_m = q_m - \Lambda \sigma_R, \bar{y} = \frac{\bar{q}_m}{\Lambda} = y - \sigma_R. \quad (30)$$

The canonical one-form changes only by a total time derivative:

$$\int (p \dot{\Lambda} + q_m \dot{R}) dx = \int (\bar{p} \dot{\Lambda} + \bar{q}_m \dot{R}) dx + \frac{dJ}{dt}. \quad (31)$$

The radial momentum generator is unchanged, because

$$\bar{q}_m R' - \Lambda \bar{p}' = q_m R' - \Lambda p'. \quad (32)$$

Substituting $p = \bar{p} + FC$ and $y = \bar{y} + (FC)_R$ into the solved function gives

$$k = -\frac{\bar{p}\bar{y}}{F} + \frac{F_R - Z}{F^2} \bar{p}^2 + U + ZC^2 + FCC_R. \quad (33)$$

The final three terms are independent of momentum and can be included in the spatial potential:

$$\bar{W} = W + U + ZC^2 + FCC_R. \quad (34)$$

Thus the linear terms give an equivalent canonical description and a redefinition of the spatial coefficient. They do not supply an additional kinetic response. With fixed configuration endpoints, the total derivative does not change the bulk equations. This is a local equivalence for smooth coefficient functions; separate global or boundary degrees of freedom are outside the specified class.

From this point on, reuse p, q_m, W for the centered momenta and transformed potential. Static momenta vanish in these variables. The selected kinetic density is

$$K_g = -\frac{pq_m}{F} + \frac{F_R - Z}{F^2} \Lambda p^2. \quad (35)$$

The quadratic form is now a consequence of the common evolution requirement within the retained spatial class. An initial time-reversal-even restriction is also unnecessary up to the canonical equivalence just shown. The functions F, Z, W remain to be fixed by the unloaded vacuum and the local metric completion.

6 The unloaded vacuum selects the spatial relations

Take a stationary areal chart $x = R$, with zero shift and vanishing momenta. Introduce

$$B = \Lambda^{-2}, h = N \Lambda, N = h \sqrt{B}. \quad (36)$$

Because $R' = 1$ and $R'' = 0$, multiplying the spatial constraint by the lapse gives $N V_g = h (F B_R / 2 + ZB + W)$. Thus the static bulk action is

$$I_{g, \text{static}} = - \int dT dR h \left(\frac{F}{2} B_R + ZB + W \right). \quad (37)$$

Vary h and B independently in the vacuum region. The first variation, before integrating the δB_R term, is

$$\delta I = - \int dT dR \left[\left(\frac{F}{2} B_R + ZB + W \right) \delta h + \frac{hF}{2} \delta B_R + hZ \delta B \right]. \quad (38)$$

For compact variations, integration by parts removes the endpoint term. The two Euler equations are

$$F B_R + 2 ZB + 2 W = 0, \frac{h_R}{h} = \frac{2 Z - F_R}{F}. \quad (39)$$

In the absence of a source, require an exactly flat unloaded solution: $S = 0$, $N = 1$, $\Lambda = 1$, hence $B = h = 1$. Applying the two equations to this state requires

$$Z + W = 0, 2 Z - F_R = 0, Z = \frac{F_R}{2}, W = -\frac{F_R}{2}. \quad (40)$$

This condition fixes the unloaded state of the isolated sector. It is stronger than asymptotic flatness alone. Use the resulting coefficient functions in the equations for a nonzero source:

$$h_R = 0, F B_R + F_R (B - 1) = 0, [F (B - 1)]_R = 0. \quad (41)$$

The asymptotic normalization makes $h = 1$. Write the remaining constant as m and solve for both field functions:

$$m = \frac{F}{2} (1 - B), N^2 = B = 1 - \frac{2m}{F(R)}, q = N^{-2}. \quad (42)$$

The clock-to-distance relation has emerged before F is fixed. Through the invariant, the stationary burden is $S^2 = 1 - N^2 = 2m / F(R)$ in these units. Its radial dependence remains to be selected.

7 The mass constant and its conserved energy

The constant in the exterior field must be connected to the generator of time evolution. After the unloaded-vacuum conditions, the constraints for arbitrary F become

$$H_g = -\frac{p q_m}{F} + \frac{F_R \Lambda p^2}{2 F^2} + V_g, D_g = q_m R' - \Lambda p', \quad (43)$$

$$V_g = \frac{F R''}{\Lambda} - \frac{F R' \Lambda'}{\Lambda^2} + \frac{F_R (R')^2}{2 \Lambda} - \frac{F_R \Lambda}{2}. \quad (44)$$

Define a mass function on general spherical slices:

$$m = \frac{F}{2} \left[1 - \left(\frac{R'}{\Lambda} \right)^2 + \left(\frac{p}{F} \right)^2 \right]. \quad (45)$$

Differentiation and substitution of the two gravitational constraints give an identity valid before the field equations are imposed:

$$m' = -\frac{R'}{\Lambda} H_g - \frac{p}{F \Lambda} D_g. \quad (46)$$

Appendix B displays the expanded terms. In a vacuum region the constraints imply $m' = 0$. In a stationary areal chart this reduces to the mass in Section 6. Thus the same variable enters both the dynamical constraints and the static exterior.

The static action becomes $\int h m_R dT dR$. Its boundary variation at infinity is $\int \delta m_\infty dT$, because the asymptotic lapse is fixed at one. The canonical endpoint convention cancels it with

$$I_\partial = -\int m_\infty dt. \quad (47)$$

For a regular center and asymptotic falloffs giving a finite charge, the total Hamiltonian is therefore

$$H_{\text{total}} = \int dx (N H + u D) + m_\infty. \quad (48)$$

The lapse and shift equations set the total constraints to zero. The conserved energy is $E = m_\infty$ in geometric units. An independent state-dependent surface energy has not been added. Restoring SI units in the static exterior gives

$$E = \frac{c^4 m_\infty}{G_*}, \alpha^2 = 1 - \frac{2 G_* E}{c^4 F(r)}, S^2(r) = \frac{2 G_* E}{c^2 F(r)}. \quad (49)$$

This identifies the energy for arbitrary F . The next section selects F by local metric compatibility and fixes the overall coupling independently of any screen condition. We retain the identification of this conserved energy with the source energy in Section 2.

8 Local metric completion selects the remaining function

The field responds to physical variables, rather than to our choice of enclosing screen. To make that statement calculable in this completion, require the spherical kinetic response to arise from one smooth local three-dimensional metric law. Its quadratic coefficients contain no extra directional field or spatial-gradient dependence. A source can still establish a directional field; this requirement concerns the variables entering the kinetic coefficients.

Keep F arbitrary and use the centered momenta of Section 5. With $Z = F_R/2$ from the unloaded vacuum, the two Hamilton equations give

$$U_R = -\frac{Np}{F}, U_\Lambda = -\frac{Nq_m}{F} + \frac{NF_R\Lambda p}{F^2}. \quad (50)$$

Solve for the momenta and take the Legendre transform, as expanded in Appendix B:

$$p = -\frac{F U_R}{N}, q_m = -\frac{F U_\Lambda + F_R \Lambda U_R}{N}, \quad (51)$$

$$L_{\text{kin}} = -\frac{1}{N} \left(F U_\Lambda U_R + \frac{F_R \Lambda}{2} U_R^2 \right). \quad (52)$$

The physical radial and tangential rates of change of distance are

$$k_r = \frac{U_\Lambda}{N \Lambda}, k_t = \frac{U_R}{N R}. \quad (53)$$

They are the principal rates of the spatial metric, with the convention $K_{ij} = (\dot{h}_{ij} - L_u h_{ij}) / (2N)$. Here h_{ij} is the three-dimensional spatial metric and L_u is its Lie derivative along the shift. The scalar $h = N \Lambda$ in the spherical static calculation is a different quantity.

Divide by lapse times physical volume per unit solid angle:

$$\ell_{\text{kin}} = \frac{L_{\text{kin}}}{N \Lambda R^2} = -\frac{F}{R} k_r k_t - \frac{F_R}{2} k_t^2. \quad (54)$$

This step uses the same physical spatial measure as the matter coupling. The areal radius and radial distance coefficient are not redefined to absorb the free function.

Local coefficient comparison

A quadratic scalar of the symmetric local rate tensor, with coefficients built only from the spatial metric, has two contractions [13]:

$$\ell_{\text{quad}} = A_K \text{tr}(K^2) + B_K (\text{tr} K)^2. \quad (55)$$

In spherical symmetry the principal rates are (k_r, k_t, k_t) , so

$$\text{tr}(K^2) = k_r^2 + 2k_t^2, (\text{tr} K)^2 = k_r^2 + 4k_r k_t + 4k_t^2. \quad (56)$$

The local quadratic response therefore restricts to

$$\ell_{\text{quad}} = (A_K + B_K)k_r^2 + 4B_K k_r k_t + (2A_K + 4B_K)k_t^2. \quad (57)$$

Match this expression to the derived spherical density for arbitrary k_r and k_t :

$$A_K + B_K = 0, 4B_K = -\frac{F}{R}, 2A_K + 4B_K = -\frac{F_R}{2}. \quad (58)$$

The first two equations give $A_K = F/(4R)$ and $B_K = -F/(4R)$. Their substitution into the third gives

$$-\frac{F}{2R} = -\frac{F_R}{2}, R F_R = F. \quad (59)$$

Integrating on a connected domain with nonzero F yields

$$F(R) = \kappa R, \ell_{\text{kin}} = \frac{\kappa}{4} [\text{tr}(K^2) - (\text{tr} K)^2]. \quad (60)$$

No screen charge, exterior burden profile or orbital measurement has entered this coefficient comparison. The positive constant κ is an overall gravitational normalization.

Quadratic spherical dynamics have not been assumed again: Section 5 derived them from arbitrary smooth momentum dependence. For this compatibility test, a completion twice differentiable near zero rates suffices, because its quadratic Taylor coefficient must have the form above. Terms invisible on all spherical configurations are not classified by this test. The minimal completion used below retains the displayed quadratic term.

Spatial completion and coupling normalization

The same selection must accommodate the spatial part. For $F = \kappa R$, the unloaded-vacuum relations give $Z = \kappa/2$ and $W = -\kappa/2$. Thus

$$V = -V_g = \kappa \left[-\frac{R R''}{\Lambda} + \frac{R R' \Lambda'}{\Lambda^2} - \frac{(R')^2}{2\Lambda} + \frac{\Lambda}{2} \right]. \quad (61)$$

The scalar curvature of $d\ell^2 = \Lambda^2 dx^2 + R^2 d\Omega^2$ is

$$R_3 = \frac{2}{R^2} \left[1 - \left(\frac{R'}{\Lambda} \right)^2 - \frac{2R}{\Lambda} \left(\frac{R'}{\Lambda} \right)' \right]. \quad (62)$$

Expanding the last derivative verifies the spatial identity

$$V = \frac{\kappa}{4} \Lambda R^2 R_3. \quad (63)$$

Both kinetic and spatial terms therefore belong to the same local metric action. Appendix F displays the full three-dimensional expression and checks its common evolution with matter. It is the Einstein action in canonical form [14].

Restore SI units. Insert $F = \kappa r$ and $E = M c^2$ into the general exterior of Section 7. The weak-field expansion of the probe action in Section 12 then gives the potential and acceleration

$$\Phi = -\frac{G_* M}{\kappa r}, a_r = -\frac{G_* M}{\kappa r^2}. \quad (64)$$

The measured gravitational coupling is consequently $G = G_*/\kappa$. The full gravitational action has prefactor $\kappa/G_* = 1/G$. Absorb that single overall factor into the coupling and use measured G henceforth:

$$F(R) = R, Z = \frac{1}{2}, W = -\frac{1}{2}. \quad (65)$$

This normalization does not select a radial function or use equality between screens. It identifies the one gravitational coupling with the same measured constant used in the source law. The retained source-energy relation is $E = E_Q = M c^2$; the numerical value of G is not predicted by the invariant.

9 The selected action and spherical exterior

The selected constraints give $P_\Lambda = -RU_R/N$ and $P_R = -(\Lambda U_R + RU_\Lambda)/N$ in geometric units. Appendix B performs the Legendre transformation before setting $F=R$. Restoring SI units produces the field action

$$I_g = \frac{c^4}{G} \int dt dx \left(NV - \frac{K}{Nc^2} \right) - \int dt M c^2, \quad (66)$$

$$V = -\frac{RR''}{\Lambda} + \frac{RR'\Lambda'}{\Lambda^2} - \frac{(R')^2}{2\Lambda} + \frac{\Lambda}{2}, \quad (67)$$

$$K = RU_\Lambda U_R + \frac{\Lambda}{2} U_R^2. \quad (68)$$

The total action includes minimally coupled matter. The spatial quantity $V = -V_g$ is dimensionless, whereas K has speed-squared units when t is measured in seconds. The expression is mathematically the spherical Einstein action [1–3,14]. Its selection used common evolution, the flat unloaded vacuum and local metric compatibility in the specified action class.

To exhibit the exterior equation directly, set $x=R=r$, $u=0$, $B=\Lambda^{-2}$ and $h=N/\sqrt{B}$. The static vacuum bulk action is

$$I_{g,\text{static}} = -\frac{c^4}{2G} \int dT dr h (rB' + B - 1). \quad (69)$$

Vary h and B independently, with compact variations. Integration by parts in the $\delta B'$ term gives

$$rB' + B - 1 = 0, h' = 0. \quad (70)$$

The first equation is $[r(1-B)]' = 0$. The asymptotic clock normalization gives $h=1$ and hence $N^2=B$. Through the invariant, $S^2=c^2(1-B)$, so

$$\frac{d}{dr} \left(\frac{rS^2}{2} \right) = 0. \quad (71)$$

Exterior screen constancy has followed from the field variation. It has not been imposed as a condition selecting the action.

The exterior source readout

The conserved energy in Section 7 fixes the integration constant. With measured G and $E = M c^2$,

$$\alpha^2 = 1 - \frac{2GM}{r c^2}, q = \alpha^{-2}, S^2 = \frac{2GM}{r}. \quad (72)$$

Leaving the source normalization explicit,

$$S^2(r) = \frac{Q_\Omega^2}{12\pi V_\Omega r} = \frac{Q_\Omega^2}{16\pi^2 R_0^3 r}. \quad (73)$$

Now define the spherical screen readout and its reference volume:

$$Q_0(r) = 4\pi r^2 S(r), V_{\text{ref}}(r) = \frac{4\pi r^3}{3}. \quad (74)$$

The solved field, with the same coupling and source-energy normalization, gives

$$\frac{Q_0(r)^2}{24\pi V_{\text{ref}}(r)} = \frac{r S^2(r)}{2} = GM = \frac{Q_\Omega^2}{24\pi V_\Omega}. \quad (75)$$

The radial constancy is a field-equation result; equality to the source charge uses the retained common coupling and mass-energy identification. The radius-dependent Q_0 is a screen readout, not a fixed volumetric discharge transported through every sphere.

The resulting interval is

$$c^2 d\tau^2 = \left(1 - \frac{2\mu}{r c^2}\right) c^2 dT^2 - \frac{dr^2}{1 - 2\mu/(r c^2)} - r^2 d\Omega^2. \quad (76)$$

This is the Schwarzschild exterior. Spherical radial density variations inside the source do not alter this vacuum solution at fixed total mass. Within matter, density and pressure enter the equations and h need not equal one. The exterior formula is not imposed throughout a material interior.

The invariant connects burden to clock progression, common evolution fixes the spherical kinetic form, and local metric compatibility fixes its remaining radial function. Source energy supplies the strength of the exterior field. A Painlevé–Gullstrand representation follows afterward in Appendix A; it is not an input to this selection.

10 The conserved energy and its static limit

In this section $m(r)$ denotes mass in kilograms. The selected spatial measure defines

$$m(r) = \frac{c^2 r}{2G} \left(1 - \frac{1}{q(r)} \right), m(\infty) = M. \quad (77)$$

The boundary normalization must also agree with time evolution. Appendix B performs the Legendre transformation of the selected field action. If H and D denote its total energy and radial momentum constraint densities in physical units, the Hamiltonian has the form

$$H_{\text{total}} = \int dx (N H + u D) + M c^2. \quad (78)$$

Variation of N and u imposes $H=0$ and $D=0$. The isolated system's time-evolution energy, normalized to t at infinity, is consequently

$$E_{\text{total}} = M c^2 = \frac{c^2 Q_\Omega^2}{24 \pi G V_\Omega}. \quad (79)$$

We can check its connection to the previous static energy before imposing the static equations. Let $\varepsilon(r)$ be matter energy per local proper volume, measured at rest. In a static chart, the momentum-dependent gravitational terms vanish. The spatial constraint contributes $-c^2 h dm/dr$, while the matter contribution is the proper shell energy multiplied by its lapse:

$$N H_m = N 4 \pi r^2 \Lambda \varepsilon = 4 \pi r^2 h \varepsilon. \quad (80)$$

The complete static Hamiltonian is thus

$$H_{\text{total,static}} = M c^2 + \int_0^\infty \left(-c^2 h \frac{dm}{dr} + 4 \pi r^2 h \varepsilon \right) dr. \quad (81)$$

Integration by parts yields $-c^2 \int h dm/dr dr = -c^2 [hm]_0^\infty + c^2 \int m dh/dr dr$. Regularity gives $m(0)=0$; the exterior endpoint gives $h(\infty)m(\infty)=M$. The endpoint cancels the explicit $M c^2$, leaving exactly

$$E_{\text{static}} = \int_0^\infty \left(c^2 m \frac{dh}{dr} + 4 \pi r^2 h \varepsilon \right) dr. \quad (82)$$

This equality is an identity for the specified static configurations and endpoints, before requiring the field equations. On a static solution, the energy constraint gives $dm/dr = 4 \pi r^2 \varepsilon / c^2$. Its substitution makes the integrand $c^2 d(hm)/dr$, so

$$E_{\text{static}} = c^2 [hm]_0^\infty = M c^2 = E_Q. \quad (83)$$

The energy bridge therefore holds for the full static matter-plus-field system. It includes the gravitational contribution; the proper material integral $\int 4 \pi r^2 \Lambda \varepsilon dr$ need not equal $M c^2$. A complete dynamical fluid action must retain its conserved-current variables. The static matter density used here is its Hamiltonian restriction, not a replacement for those variables. Appendix D checks a moving pressureless source.

11 The normalized readout and light propagation

The solved exterior field defines a spherical screen readout at every exterior radius:

$$Q_0(r) = 4\pi r^2 S(r) = 4\pi \sqrt{2\mu r^3}, V_{\text{ref}}(r) = \frac{4\pi r^3}{3}. \quad (84)$$

Squaring gives $Q_0^2/V_{\text{ref}} = 24\pi\mu$, the source-screen equality obtained from the solved exterior in Section 9. The fixed-domain Q_Ω and radius-dependent $Q_0(r)$ must be distinguished. In particular, Q_0 varies as $r^{3/2}$ even though μ stays fixed.

For the held-observer normalization, use $Q_* = Q_0/\alpha$. Substituting both solved functions gives the expression from the earlier algebraic tests:

$$Q_*(r) = \frac{4\pi r^2 S(r)}{\sqrt{1 - S(r)^2/c^2}} = 4\pi r^{3/2} \sqrt{\frac{2\mu}{1 - 2\mu/(rc^2)}}. \quad (85)$$

This changes the clock normalization of the readout. It does not increase the source energy. To invert it, put $z = Q_*/(4\pi r^2 c)$ and $y = S^2/c^2$. Then

$$z^2 = \frac{y}{1-y}, z^2(1-y) = y, y = \frac{z^2}{1+z^2}. \quad (86)$$

$$\alpha^2 = 1 - y = \frac{1}{1+z^2}, q = \alpha^{-2} = 1 + z^2. \quad (87)$$

The final relation for q uses the field solution in Section 9. The normalized readout agrees with that solution; a held-clock measurement alone would not determine a spatial coefficient.

For radial light, set $d\tau = 0$ and $d\Omega = 0$ in the solved interval. The two radial terms must cancel:

$$0 = \alpha^2 c^2 dT^2 - \alpha^{-2} dr^2 \Rightarrow \left| \frac{dr}{dT} \right| = c \alpha^2. \quad (88)$$

The radial ruler and held clock use $d\ell = |dr|/\alpha$ and $d\tau_{\text{hold}} = \alpha dT$. Therefore

$$\frac{d\ell}{dT} = c\alpha, \frac{d\ell}{d\tau_{\text{hold}}} = \frac{c\alpha}{\alpha} = c. \quad (89)$$

The common clock factor cancels in the local speed measurement. Signal travel timed against T retains both the clock factor and radial spatial response. These are different operational measurements of the same solved geometry.

12 The test body action and conserved quantities

Use a test body of rest mass m_p , with negligible backreaction and fixed trajectory endpoints. Adopt its minimally coupled proper-time action:

$$I_p = -m_p c^2 \int d\tau = \int L dT, L = -m_p c^2 \beta, \delta I_p = 0. \quad (90)$$

Spherical symmetry places the orbit in an equatorial plane. In Sections 12–13 only, a dot denotes d/dT . The invariant gives

$$\beta = \sqrt{\alpha^2 - \frac{q\dot{r}^2 + r^2\dot{\phi}^2}{c^2}}. \quad (91)$$

Differentiate L with respect to each coordinate velocity:

$$p_r = \frac{m_p q \dot{r}}{\beta}, p_\phi = \frac{m_p r^2 \dot{\phi}}{\beta} = m_p j, j = r^2 \frac{d\phi}{d\tau}. \quad (92)$$

The Lagrangian has no explicit ϕ dependence, so j is conserved. It has no explicit T dependence, so its Hamiltonian is conserved. Using $q\dot{r}^2 + r^2\dot{\phi}^2 = c^2(\alpha^2 - \beta^2)$ in the Legendre transform gives

$$E_p = p_r \dot{r} + p_\phi \dot{\phi} - L = \frac{m_p}{\beta} [q\dot{r}^2 + r^2\dot{\phi}^2 + c^2\beta^2] = \frac{m_p c^2 \alpha^2}{\beta}. \quad (93)$$

$$\varepsilon \equiv \frac{E_p}{m_p c^2} = \frac{\alpha^2}{\beta}, \frac{dT}{d\tau} = \frac{\varepsilon}{\alpha^2}. \quad (94)$$

The speed measured by a held local observer is $v_{\text{loc}}^2 = (q\dot{r}^2 + r^2\dot{\phi}^2)/\alpha^2$. Substitution into β^2 gives $\beta = \alpha/\gamma_{\text{loc}}$, where $\gamma_{\text{loc}} = (1 - v_{\text{loc}}^2/c^2)^{-1/2}$. Thus $\varepsilon = \alpha\gamma_{\text{loc}}$. Marginal radial escape has $\varepsilon = 1$, since it reaches infinity at rest:

$$\alpha\gamma_{\text{esc}} = 1 \Rightarrow 1 - \frac{v_{\text{esc}}^2}{c^2} = \alpha^2 \Rightarrow v_{\text{esc}}^2 = S^2. \quad (95)$$

This is the exact local escape-speed relation in the stationary spherical exterior. For slow motion and $S^2/c^2 \ll 1$, expanding the action gives $L = -m_p c^2 + m_p(v^2 + S^2)/2$ to leading order, where $v^2 = \dot{r}^2 + r^2\dot{\phi}^2$ at this order. The resulting weak-field acceleration is $a = \nabla S^2/2$. For perihelion we keep the full action and spatial response instead of stopping at this limit.

13 The exact exterior orbit equation

Divide the solved interval by $d\tau^2$, then substitute the conserved energy and angular momentum:

$$c^2 = \alpha^2 c^2 \left(\frac{dT}{d\tau} \right)^2 - \alpha^{-2} \left(\frac{dr}{d\tau} \right)^2 - r^2 \left(\frac{d\phi}{d\tau} \right)^2. \quad (96)$$

$$\alpha^2 c^2 = c^2 \varepsilon^2 - \left(\frac{dr}{d\tau} \right)^2 - \frac{\alpha^2 j^2}{r^2}. \quad (97)$$

Isolate the radial derivative and use $\alpha^2 = 1 - 2\mu/(rc^2)$:

$$\left(\frac{dr}{d\tau} \right)^2 = c^2 \varepsilon^2 - \left(1 - \frac{2\mu}{rc^2} \right) \left(c^2 + \frac{j^2}{r^2} \right). \quad (98)$$

For an orbit with $j \neq 0$, define $w = 1/r$. A prime in the remaining orbital derivation denotes $d/d\phi$. The chain rule and $d\phi/d\tau = j/r^2$ give

$$\frac{dr}{d\tau} = -\frac{1}{w^2} \frac{dw}{d\phi} \frac{d\phi}{d\tau} = -j w'. \quad (99)$$

Insert this into the radial first integral. Expanding the product before division by j^2 displays the cubic term:

$$j^2 (w')^2 = c^2 (\varepsilon^2 - 1) + 2\mu w - j^2 w^2 + \frac{2\mu j^2 w^3}{c^2}. \quad (100)$$

$$(w')^2 = \frac{c^2 (\varepsilon^2 - 1)}{j^2} + \frac{2\mu w}{j^2} - w^2 + \frac{2\mu w^3}{c^2}. \quad (101)$$

Differentiate with respect to ϕ , holding ε and j constant:

$$2w'w'' = \frac{2\mu}{j^2} w' - 2w w' + \frac{6\mu w^2}{c^2} w'. \quad (102)$$

Division by $2w'$ away from a turning point, followed by continuous extension to each turning point, yields

$$\frac{d^2 w}{d\phi^2} + w = \frac{\mu}{j^2} + \frac{3\mu w^2}{c^2}. \quad (103)$$

The extra term originates in the product of the source factor $2\mu w/c^2$ with the angular term $j^2 w^2$ in the radial energy equation. No orbital correction has been added afterward. The dimensions agree: each term has units of inverse length, since j has units of area per time.

14 From the orbit equation to perihelion advance

Let $p_0 = j^2 / \mu$ and $\lambda = \mu / (p_0 c^2) \ll 1$. At zeroth order, the orbit equation has the Kepler solution

$$w_N = \frac{1 + e \cos \varphi}{p_0}. \quad (104)$$

Write $w = w_N + \delta w$. To first order in λ , replace w by w_N only in the already small term $3\mu w^2 / c^2$. Expanding the square and using $\cos^2 \varphi = (1 + \cos 2\varphi) / 2$ gives

$$\delta w'' + \delta w = \frac{3\mu}{c^2 p_0^2} \left(1 + \frac{e^2}{2} + 2e \cos \varphi + \frac{e^2}{2} \cos 2\varphi \right). \quad (105)$$

The constant and twice-frequency forcing give bounded corrections. The forcing at the unperturbed radial frequency produces the accumulating phase change. Indeed, $(d^2 / d\varphi^2 + 1)(\varphi \sin \varphi) = 2 \cos \varphi$, so

$$\delta w_{\text{res}} = \frac{3\mu e}{c^2 p_0^2} \varphi \sin \varphi. \quad (106)$$

Compare this term with the expansion of a slowly shifted radial phase:

$$\cos[(1 - \delta)\varphi] = \cos \varphi + \delta \varphi \sin \varphi + O(\delta^2), \quad \delta = \frac{3\mu}{p_0 c^2}. \quad (107)$$

Successive perihelia occur after the radial phase advances by 2π . Their azimuthal separation is $2\pi / (1 - \delta)$. Subtract one full turn and expand to first order:

$$\Delta \varpi = \frac{2\pi}{1 - \delta} - 2\pi = \frac{6\pi\mu}{p_0 c^2} + O(\lambda^2). \quad (108)$$

At the retained order, $p_0 = a(1 - e^2)$ for semimajor axis a and eccentricity e . Restoring $\mu = GM$ gives

$$\Delta \varpi = \frac{6\pi GM}{a(1 - e^2)c^2} + O(\lambda^2). \quad (109)$$

The advance is positive. It measures a mismatch between radial and azimuthal periods; a relabeling of time does not remove the angle between successive perihelia. Substituting the SET source normalization leaves the source variables explicit:

$$\Delta \varpi = \frac{Q_\Omega^2}{4V_\Omega p_0 c^2} + O(\lambda^2) = \frac{3Q_\Omega^2}{16\pi R_0^3 p_0 c^2} + O(\lambda^2). \quad (110)$$

The source reference radius R_0 and orbital scale p_0 have distinct roles. A new choice of R_0 requires its corresponding Q_Ω at the same μ . Consequently the calculated advance is independent of an arbitrary change of source reference domain.

15 Mercury and the exact orbital check

For a Mercury-scale example, use $c=299792458$ m/s and $\mu_{\odot}=1.3271244004127942 \times 10^{20}$ m³/s² [5]. JPL's approximate J2000 elements give $a=0.38709927$ au and $e=0.20563593$ [6]. With 1 au=149597870700 m,

$$a=5.790922654 \times 10^{10} \text{ m}, p_N=a(1-e^2)=5.546046913 \times 10^{10} \text{ m}. \quad (111)$$

Quantity	Calculated value
Leading advance per orbit	5.018660440e-07 rad
Leading advance per orbit	0.103517302 arcsec
Leading advance per Julian century	42.98070 arcsec
Exact spherical advance per Julian century	42.98071 arcsec

The conversion uses 87.969 days per orbit [7] and 36525 days per Julian century. Extra digits distinguish the numerical calculations, not their observational precision. The result is about 43 arcseconds per century, the isolated spherical relativistic contribution. It is the Schwarzschild result; Mercury's total precession also includes planetary perturbations and solar multipoles.

To check the approximation against the exact equation, define turning radii $r_p=p/(1+e)$, $r_a=p/(1-e)$ and $\ell_g=\mu/c^2$. The turning-point parameter p equals p_0 only at leading order. Evaluate the radial first integral at r_p and r_a , where $dr/d\tau=0$, and equate the two values of ε^2 . Solving first for j^2 and then for ε^2 gives

$$j^2=\frac{\mu p^2}{p-(3+e^2)\ell_g}, \varepsilon^2=\frac{(p-2\ell_g)^2-4e^2\ell_g^2}{p[p-(3+e^2)\ell_g]}. \quad (112)$$

Use $w=(1+e\cos\psi)/p$ to label the radial phase. Then $dw/d\psi=-e\sin\psi/p$. Substitution of these constants into the first integral factors its radial dependence as

$$(w')^2=\frac{e^2\sin^2\psi}{p^2}\left[1-\frac{2\ell_g}{p}(3+e\cos\psi)\right]. \quad (113)$$

Canceling the turning-point factor with $(dw/d\psi)^2$ gives

$$\frac{d\varphi}{d\psi}=\left[1-\frac{2\ell_g}{p}(3+e\cos\psi)\right]^{-1/2}. \quad (114)$$

The exact excess angle follows by integrating one radial cycle:

$$\Delta\varphi_{\text{exact}}=2\int_0^\pi\left[\left[1-\frac{2\ell_g}{p}(3+e\cos\psi)\right]^{-1/2}-1\right]d\psi. \quad (115)$$

This applies to the nonplunging bound branch $p>(6+2e)\ell_g$. Expansion of the integrand gives $6\pi\ell_g/p$ at first order. The exact Mercury benchmark sets $p=p_N$ and uses the same eccentricity. Appendix E compares this quadrature with two independent orbit integrations.

Appendix A Free fall and the source volume

The volume correspondence investigated before this action can now be derived from the solution. A radial observer released from rest at infinity has $\varepsilon = 1$ and $j = 0$. The conserved energy and radial first integral give

$$\frac{dT}{d\tau_{\text{ff}}} = \alpha^{-2}, \frac{dr}{d\tau_{\text{ff}}} = -S. \quad (116)$$

Introduce a new time coordinate by

$$dt_{\text{ff}} = dT + \frac{S}{\alpha^2 c^2} dr. \quad (117)$$

The coefficient is a function of r only, so this differential is integrable. Along the falling observer, $dt_{\text{ff}}/d\tau_{\text{ff}} = \alpha^{-2} - S^2/(\alpha^2 c^2) = 1$. Substituting for dT in the static interval and completing the square gives [4]

$$c^2 d\tau^2 = c^2 dt_{\text{ff}}^2 - (dr + S dt_{\text{ff}})^2 - r^2 d\Omega^2. \quad (118)$$

This is the ingoing Painlevé–Gullstrand representation. On its constant-time slices the radial measure is $d\chi = dr = \alpha d\ell$ for outward increments. The screen normalization defines

$$V_Q(r) = \frac{Q_0(r)^2}{24\pi\mu} = \frac{4\pi r^3}{3}, dV_Q = 4\pi r^2 dr = dV_{\text{ff}}. \quad (119)$$

This is an equality of shell volumes in the exterior free-fall slicing. It does not extend a volume formula through an unspecified material interior. Along the inward trajectory, differentiate both readouts:

$$\frac{dV_Q}{d\tau_{\text{ff}}} = 4\pi r^2(-S) = -Q_0, \frac{dQ_0}{d\tau_{\text{ff}}} = \frac{3Q_0}{2r}(-S) = -12\pi\mu. \quad (120)$$

These derivatives describe what the falling observer samples. They do not assert that a fixed discharge is transported through all spheres at speed S . The fixed source strength remains μ .

The physical-gradient balance also follows. Define the outward magnitude J using the free-fall radial measure; using the solved profile gives

$$J = -\frac{1}{2} \frac{dS^2}{d\chi} = \frac{\mu}{r^2}, \oint_{\Sigma_r} J dA = 4\pi\mu = \frac{Q_\Omega^2}{6V_\Omega}. \quad (121)$$

The held support acceleration is $a_{\text{hold}} = c^2 d \ln \alpha / d\ell$. Because $S^2 = c^2(1 - \alpha^2)$ and $d\chi = \alpha d\ell$,

$$J = c^2 \frac{d\alpha}{d\ell} = \alpha a_{\text{hold}}. \quad (122)$$

The previously retained gradient normalization is therefore the redshifted support-acceleration flux of this action's solution. The action selection and exterior solution precede this coordinate interpretation. The volume equality follows from that solution.

Appendix B The common geometry and constraint calculation

This appendix supplies the intermediate steps behind Sections 5 and 7. Use $G_*=c=1$ before coupling normalization, and $G=c=1$ for the final normalized action. In a locally unloaded time-radial plane the interval is $d\tau^2 = dT^2 - dX^2$. Parameterize an initial spatial line by $X(x)$, with $dX/dx = \Lambda > 0$. A normal advance of size $sN_1(x)$ produces the line $(T, X) = (sN_1(x), X(x))$. Its future unit normal is

$$n = \frac{(\Lambda, sN_1')}{\sqrt{\Lambda^2 - s^2(N_1')^2}}. \quad (123)$$

Advance a second time by sN_2 along this normal, and subtract the result with the order reversed. The leading spatial difference is

$$\Delta X = \frac{s^2(N_2N_1' - N_1N_2')}{\Lambda} + O(s^3), \quad (124)$$

$$\Delta x = \frac{\Delta X}{\Lambda} = \frac{s^2(N_2N_1' - N_1N_2')}{\Lambda^2} + O(s^3). \quad (125)$$

The normal component vanishes in the commutator limit. The sign depends on the order convention; the coefficient is the inverse radial spatial metric. Requiring canonical matter-field evolution to represent this geometry gives the bracket in Section 4. This uses the common Lorentzian interpretation and freely relabelable slices, rather than following from a scalar numerical bound alone.

For the functional calculation, take compactly supported lapse smearings so endpoint terms vanish. Integration by parts gives

$$\int N V_g dx = \int dx \left\{ N \left[\frac{(Z - F_R)(R')^2}{\Lambda} + W \Lambda \right] - \frac{N' F R'}{\Lambda} \right\}. \quad (126)$$

The functional derivative with respect to Λ has an N' coefficient A_c . The derivative with respect to R has coefficients B_c for N' and C_c for N'' :

$$A_c = \frac{F R'}{\Lambda^2}, B_c = \frac{2(F_R - Z)R'}{\Lambda} - \frac{F \Lambda'}{\Lambda^2}, C_c = \frac{F}{\Lambda}. \quad (127)$$

Terms proportional to $N_1 N_2$ cancel in the antisymmetrization of the Poisson bracket. This includes the terms involving W . The derivative terms retain information about F and Z .

Evaluation for arbitrary momentum dependence

The momentum density is $K_g = \Lambda k(R, p, y)$, with $y = q_m / \Lambda$. Its two momentum derivatives are

$$K_p = \Lambda k_p, K_{q_m} = k_y. \quad (128)$$

Terms with no lapse derivatives cancel in the antisymmetric Poisson bracket. One integration by parts on the second lapse derivative gives the gravitational coefficient of $\omega = N_1 N_2' - N_2 N_1'$:

$$B_g = -A_c K_p - B_c K_{q_m} + (C_c K_{q_m})'. \quad (129)$$

The coefficients A_c, B_c, C_c were obtained on the preceding page from the spatial functional. Expand the last derivative using

$$C_c' = \frac{F_R R'}{\Lambda} - \frac{F \Lambda'}{\Lambda^2}, \quad (130)$$

$$(k_y)' = k_{yR} R' + k_{yp} p' + k_{yy} y', y' = \frac{q_m'}{\Lambda} - \frac{q_m \Lambda'}{\Lambda^2}. \quad (131)$$

Before collecting terms, the coefficient is

$$B_g = -\frac{F R'}{\Lambda} k_p - \frac{2(F_R - Z) R'}{\Lambda} k_y + \frac{F \Lambda'}{\Lambda^2} k_y + E_c, \quad (132)$$

$$E_c = \left(\frac{F_R R'}{\Lambda} - \frac{F \Lambda'}{\Lambda^2} \right) k_y + \frac{F}{\Lambda} (k_{yR} R' + k_{yp} p' + k_{yy} y'). \quad (133)$$

The explicit $\Lambda' k_y$ terms cancel. Collecting the other terms yields the coefficient in Section 5:

$$B_g = \frac{1}{\Lambda} [F k_{yp} p' + F k_{yy} y' + R' (-F k_p + (2Z - F_R) k_y + F k_{yR})]. \quad (134)$$

Matter contributes D_m / Λ^2 . Mixed terms cancel because the minimal matter Hamiltonian has no spatial derivatives of the metric variables. Therefore

$$\{H[N_1], H[N_2]\} = \int dx \omega \left(B_g + \frac{D_m}{\Lambda^2} \right). \quad (135)$$

Independent coefficient comparison gives the three functional equations solved in Section 5. This calculation leaves F, Z, W free until the unloaded vacuum and local metric completion are applied. It assumes no momentum derivatives or mixed momentum-gradient operators, and retains the spatial form stated in Section 4.

Mass identity and Legendre transformation before selection

After the flat-vacuum condition fixes $Z = F_R/2$ and $W = -F_R/2$, keep F arbitrary. Differentiating the mass function in Section 7 gives

$$m' = \frac{F_R R'}{2} \left[1 - \frac{(R')^2}{\Lambda^2} - \frac{p^2}{F^2} \right] - \frac{F R' R''}{\Lambda^2} + \frac{F (R')^2 \Lambda'}{\Lambda^3} + \frac{p p'}{F}. \quad (136)$$

Now expand $-(R'/\Lambda)H_g - p D_g/(F\Lambda)$. The two $p q_m R'/(F\Lambda)$ terms cancel. Its remaining terms are exactly those above. This proves the mass identity without imposing the constraints or choosing $F = R$.

The Hamilton equations for the two configuration variables give

$$U_R = -\frac{Np}{F}, U_\Lambda = -\frac{Nq_m}{F} + \frac{NF_R \Lambda p}{F^2}. \quad (137)$$

Inverting them yields

$$p = -\frac{F U_R}{N}, q_m = -\frac{F U_\Lambda + F_R \Lambda U_R}{N}. \quad (138)$$

Use $\dot{R} = U_R + u R'$ and $\dot{\Lambda} = U_\Lambda + (u \Lambda)'$ in the Legendre transform. The shift terms reproduce $u D_g$ after one integration by parts. The velocity-dependent bulk terms reduce to

$$p U_\Lambda + q_m U_R - N K_g = -\frac{1}{N} \left(F U_\Lambda U_R + \frac{F_R \Lambda}{2} U_R^2 \right). \quad (139)$$

Thus the general bulk action at this stage is

$$I_{g,\text{bulk}} = \int dt d\mathbf{x} \left[-N V_g - \frac{F U_\Lambda U_R + (F_R \Lambda / 2) U_R^2}{N} \right]. \quad (140)$$

Setting $F = R$ after this transformation gives the kinetic and spatial terms in Section 9. In SI units, the spatial part carries c^4/G and the quadratic velocity part carries c^2/G . Section 8 independently identifies the overall coupling with measured G . The source-energy normalization uses this same constant.

Appendix C Alternative actions and the selection domain

An explicit alternative helps separate the conditions. Let L_0 be a fixed length and take

$$F(R) = R + \frac{\lambda L_0^2}{R}, Z = \frac{F_R}{2}, W = -\frac{F_R}{2}. \quad (141)$$

Use the kinetic density determined in Section 5. On a domain where $F > 0$, this action passes the spherical common-evolution bracket and admits a flat unloaded state. It therefore survives those earlier requirements. The new local metric condition tests

$$\frac{F}{R} - F_R = \frac{2\lambda L_0^2}{R^2}. \quad (142)$$

For nonzero λ this is nonzero, so the derived spherical kinetic coefficients cannot be the restriction of the metric-only quadratic response in Section 8. This is the exclusion criterion. Its exterior screen behavior can then be inspected as a consequence:

$$\alpha^2 = 1 - \frac{2G_*E}{c^4 F(r)}, \frac{\mu_{\text{screen}}}{G_*E/c^2} = \frac{r^2}{r^2 + \lambda L_0^2}. \quad (143)$$

The following mathematical example sets $G_*E/c^4 = L_0 = 1$ and $r = 4$. No measured orbit is used:

λ	α^2	$\mu_{\text{screen}} / (G_*E/c^2)$
0.00	0.500000	1.000000
0.25	0.507692	0.984615
1.00	0.529412	0.941176
-0.25	0.492063	1.015873

The deformed alternatives have radius-dependent screen ratios, but that behavior was not used to select F . They fail the preceding local coefficient comparison. The surviving $F = \kappa R$ rescales the full gravitational action; Section 8 absorbs that factor into the independently measured coupling.

The metric-only restriction is stronger than independence from an arbitrary enclosing boundary. An actual source gradient can define a direction without identifying any chosen screen. The next page makes that distinction explicit.

The physical scope of the local completion

Allow a physical unit direction n^i . A quadratic response can then include

$$\ell = \frac{a_{\text{loc}}}{4} [\text{tr}(K^2) - (\text{tr} K)^2] + \frac{d_{\text{loc}}}{4} [\text{tr} K - n^i K_{ij} n^j]^2, \quad (144)$$

$$a_{\text{loc}} = \frac{F}{R}, d_{\text{loc}} = \frac{1}{2} \left(\frac{F}{R} - F_R \right). \quad (145)$$

For a radial n^i and principal rates (k_r, k_t, k_t) , the last square is $4k_t^2$. Substitution gives

$$\ell = -\frac{F}{R} k_r k_t - \frac{F_R}{2} k_t^2. \quad (146)$$

This reconstructs the arbitrary- F spherical kinetic density. It remains a scalar when both the rate tensor and the physical direction are rotated. Thus rotation covariance alone is insufficient if such directional data are allowed. This is a kinetic counterexample to an unrestricted uniqueness claim; a full alternative action and its constraints have not been established by it.

The direction could arise from another field or from permitted spatial gradients. Both change the kinetic class retained here. Likewise, higher spatial derivatives, lapse-gradient interactions, nonlocal terms and independent clock fields lie outside the stated construction. Canonical shifts and endpoint terms remain equivalent descriptions when they preserve the physical variables and boundary conditions.

The invariant supplies the burden-to-clock relation. Common operational geometry restricts field evolution. Local metric compatibility selects the radial function, while measured coupling and source energy fix its normalization. The source law and invariant alone do not establish every one of these dynamical premises.

The resulting spherical completion is mathematically Einstein gravity. If its minimal action and matter coupling were applied universally, they would give GR dynamics. This calculation does not derive SET's separate galaxy-area response or exclude additional dynamics that vanish in spherical symmetry. Its claim is the stated spherical derivation through the exterior orbit.

Appendix D A time dependent source check

A moving spherical source also tests the energy accounting. Use $G=c=1$ and a homogeneous pressureless ball, initially at rest, with surface radius one and mass $M_0=0.05$. For the comoving label $0 \leq x \leq 1$, set

$$R(t, x) = a(t)x, \Lambda = \frac{a(t)}{\sqrt{1 - 2M_0x^2}}, N=1, u=0, \quad (147)$$

$$\rho = \frac{3M_0}{4\pi a^3}, \ddot{a} = -\frac{M_0}{a^2}, a(0)=1, \dot{a}(0)=0. \quad (148)$$

Multiplying the evolution equation by $2\dot{a}$ and integrating from rest yields $\dot{a}^2 = 2M_0(1/a - 1)$. The momenta from Appendix B are $P_A = -a\dot{a}x^2$ and $P_R = -2a\dot{a}x/\sqrt{1 - 2M_0x^2}$. Insert them into the mass function:

$$m(t, x) = \frac{ax}{2} [2M_0x^2 + \dot{a}^2x^2] = M_0x^3. \quad (149)$$

Every comoving shell therefore keeps its mass in this pressureless example. The surface matches a vacuum exterior of mass M_0 . The numerical integration solves $\ddot{a} = -M_0/a^2$ directly; the first integral and the mass identity are checked on its output.

R/R_0	Density ratio	S_{ext}/c	$Q_{\text{ref}}/Q_{\text{initial}}$	$E/(M_0c^2)$
1.000	1	0.31623	1.000000	1.0000000000
0.750	2.3704	0.36515	0.649519	1.0000000000
0.500	8	0.44721	0.353553	1.0000000000
0.250	64	0.63246	0.125000	1.0000000000
0.125	512	0.89443	0.044194	1.0000000000

The table evaluates S_{ext} from the stationary exterior just outside the moving surface. It is not the comoving lapse readout and is not the surface infall speed. The reference radius changes during collapse, so $Q_{\text{ref}} \propto R^{3/2}$ changes while

$$\frac{Q_{\text{ref}}^2}{V_{\text{ref}}} = 24\pi G M_0, E_{\text{total}} = M_0c^2 \quad (150)$$

remains fixed, with physical units restored in this line. The proper rest mass of this ball is 0.05158596920 in the geometrized units, whereas its total gravitational mass is 0.05. Their difference illustrates the binding-energy distinction used in Section 2.

For a comoving perfect fluid with pressure p_f and no heat flux, the corresponding local mass balance is [9]

$$D_\tau(m c^2) = -4\pi R^2 p_f D_\tau R. \quad (151)$$

Here D_τ differentiates along the fluid's proper time. An internal shell can exchange energy by pressure work; the dust example makes that term zero. This is a mathematical check of the spherical completion and source-energy accounting, rather than an observational fit or a nonspherical radiation test.

Appendix E Verification of the calculation

The orbit calculation was checked in three independent ways: the exact radial-phase quadrature in Section 15, direct integration of the second-order Binet equation, and proper-time integration of the radial geodesic equation. The latter uses $d^2r/d\tau^2 = -\mu/r^2 + j^2/r^3 - 3\mu j^2/(c^2 r^4)$ together with $d\phi/d\tau = j/r^2$ and $dT/d\tau = \varepsilon/\alpha^2$. Each route starts from the same turning radii and conserved quantities.

p/ℓ_g	e	Exact advance rad	Largest route difference rad
1000	0.2	0.0189349966	1.76e-15
100	0.6	0.1976235573	5.91e-15
20	0.2	1.2278075089	8.44e-15
8	0.5	6.9699200547	5.33e-15

The largest absolute residual of the dimensionless capacity invariant across these orbit integrations was 4.44e-16. The equation was evaluated along each integrated trajectory rather than imposed as a replacement evolution equation.

The action checks also evaluate its static reduction before solving the field equations, its Legendre transformation, the source-normalization substitutions and the factorization used in the exact orbital integral. Exact rational substitutions and independent functional variations check the canonical identities. The analytic derivations establish the identities; finite substitutions check their implementation.

The time-dependent source check sampled 235 space-time points and tested both constraints, the mass and all four gravitational Hamilton equations. Its maximum normalized residual was 2.85e-12. Three static fluid configurations gave a maximum relative energy mismatch of 6.66e-16. Four static profiles not required to solve the field equations gave a maximum absolute energy-identity residual of 1.33e-15 in the dimensionless test units.

The earlier quadratic-coefficient calculation checked 1299 exact rational substitutions, including the general bracket, static equations, mass identity and Legendre transform. Twelve independent functional bracket evaluations gave a maximum residual of 4.16e-17. Twenty-four finite-difference functional-derivative checks gave a maximum discrepancy of 2.81e-12. Closed non-Einstein alternatives passed the bracket, while a deliberately inconsistent coefficient choice failed it by 0.01515 on the same test fields.

The earlier normal-advance check also used six step sizes. Reducing the step from 0.1 to 0.003125 reduced the scaled spatial discrepancy from 7.07×10^{-4} to 6.89×10^{-7} . These are convergence and implementation checks of the analytic argument.

These checks preserve the energy and orbital results while the argument selecting the action is extended. The next page records the additional tests of arbitrary momentum dependence.

Verification of the unrestricted momentum calculation

The unrestricted-momentum calculation checked the local bracket identity, the general functional-equation solution and the canonical momentum shift with 1440 exact rational substitutions. It then evaluated the full smeared Poisson bracket from independent functional derivatives of the Hamiltonian on periodic grids with 128, 256 and 512 points.

Across 21 smeared-bracket evaluations, the compact formula and full functional calculation agreed to $5.55\text{e-}17$. Forty-two independent finite-difference functional-derivative checks gave a maximum discrepancy of $1.89\text{e-}12$. The following residuals use the 512-point grid and dimensionless test variables; each alteration is added to the selected function unless otherwise stated.

Momentum response	Absolute bracket residual
Selected kinetic form	0.000e+00
Equivalent canonical shift	2.082e-17
Added $0.4(p/R)^4$	6.468e-05
Added $0.13y^2$	8.442e-02
Added $0.3(p/R)^2y^2$	4.015e-03
Added $0.3[1 - \cos(p/R)]^2$	1.171e-05
Independent sixth degree trial	5.852e-02

The unmodified kinetic form and its canonical-shift representative satisfy the same common evolution bracket. Added fourth-power, mixed and nonpolynomial momentum terms fail that bracket on the tested configurations. Refining the grid leaves their nonzero residuals unchanged to numerical precision. The analytic functional equations in Section 5, rather than the finite list of trial terms, establish the restriction on arbitrary smooth momentum dependence.

The canonical-shift test is important because a different expression need not represent different physics. Its action changes by the explicitly derived total derivative and spatial-potential redefinition, and its momentum generator is unchanged. The test therefore retains an equivalent description while rejecting the deliberately altered kinetic responses within the same spatial class.

The reported residuals are numerical consistency errors. They are not uncertainties in measured gravity or evidence distinguishing this spherical completion from GR. Mercury remains the spherical relativistic benchmark in Section 15. The action selection remains conditional on the common geometric evolution, the specified field content and spatial structure, and the local metric completion.

Capacity universality has a calculable role in this construction: requiring the field to realize the same evolution geometry as material events determines its kinetic form. The local coefficient calculation in Section 8 then selects the radial response. Source normalization, the exterior orbit equation and the perihelion advance follow in the order displayed. The additional checks below test that new selection step.

Verification of local completion and source independence

The kinetic compatibility calculation was checked with 270 exact rational substitutions and 100 rotation and reconstruction trials, including the extra-direction counterexample in Appendix C. The latter tests had a maximum numerical error of 2.67×10^{-15} . The spatial-curvature, kinetic and Legendre identities were checked with a further 240 exact substitutions. Appendix F gives the three-dimensional evolution calculation and its numerical verification.

A separate source check used $G=c=1$, fixed gravitational mass $M=0.05$, and observation radius $r=8$. For source radii $a=1,2,4$, prescribe two energy-density profiles per radius:

$$\rho(r) = \rho_c \left[1 - \gamma \left(\frac{r}{a} \right)^2 \right], \rho_c = \frac{M}{4\pi a^3 (1/3 - \gamma/5)}, \gamma = 0, 0.8. \quad (152)$$

Here ρ is local energy density in these units. The constraint gives $m(r) = 4\pi \int_0^r \rho(s) s^2 ds$ and $B = 1 - 2m/r$. Integrate the exterior equation $B' = (1 - B)/r$ outward from the source. For the interior, integrate the pressure and lapse equations inward from $p(a) = 0$ and the exterior value of $N(a)$:

$$\frac{d \ln N}{dr} = \frac{m + 4\pi r^3 p}{r^2 B}, \frac{dp}{dr} = -(\rho + p) \frac{d \ln N}{dr}. \quad (153)$$

a	Central lapse uniform	Central lapse concentrated	$S(8)$
1	0.9230249471	0.9112236076	0.1118033989
2	0.9620191517	0.9561856651	0.1118033989
4	0.9811313244	0.9782308206	0.1118033989

The central lapse varies with source radius and concentration, while the exterior burden at the same observation radius agrees. All six models give $r S^2/2 = 0.05$, with maximum numerical error 1.60×10^{-15} . The uniform-source central lapse agrees with its analytic solution to 1.11×10^{-16} . The numerical center is evaluated at $r/a = 10^{-6}$. These are static mathematical source models; no equation of state, causal sound speed or stability test is inferred.

Appendix F Three dimensional local completion

The local coefficient comparison needs a compatible spatial and evolution law. Let h_{ij} be the physical spatial metric, $\sqrt{\det h}$ its volume density, and $K_{ij} = (\dot{h}_{ij} - L_u h_{ij}) / (2N)$ its local rate tensor. With t in seconds, K_{ij} has rate units. The minimal action whose spherical restriction was obtained in Section 8 is [14]

$$I_g = \frac{\kappa c^4}{16\pi G_*} \int dt d^3x N \sqrt{\det h} \left[R_3 + \frac{K_{ij} K^{ij} - (\text{tr } K)^2}{c^2} \right] + I_\partial. \quad (154)$$

The angular integral is 4π . Its spherical kinetic density is $\kappa N \Lambda R^2 [k_r^2 + 2k_t^2 - (k_r + 2k_t)^2] / 4$, exactly the selected expression. The spatial identity in Section 8 gives the other term. Absorbing $\kappa / G_* = 1 / G$ yields the normalized action used in Section 9.

The spatial class retains local metric terms of total derivative order at most two. Within that class, scalar curvature supplies the nonconstant spatial term up to a divergence. A constant term is set to zero by the exactly flat unloaded vacuum. Curvature squares or derivative-dependent kinetic coefficients are not included. The direct spherical identities establish the completion used here without claiming a classification beyond this class.

To test common evolution, use geometrized units and write the Hamiltonian with independent kinetic and spatial coefficients:

$$H_g[N] = \int d^3x N \left[\frac{a}{\sqrt{\det h}} (\pi^{ij} \pi_{ij} - \lambda_H \pi^2) - b \sqrt{\det h} R_3 \right]. \quad (155)$$

The momentum π^{ij} is a symmetric tensor density of weight one and $\pi = h_{ij} \pi^{ij}$. The symbols a, b in this appendix are action coefficients, not a source radius or orbital element. A minimally coupled massless scalar shares the same physical metric and lapse:

$$H_m[N] = \frac{1}{2} \int d^3x N \left[\frac{\Pi^2}{\sqrt{\det h}} + \sqrt{\det h} h^{ij} \partial_i \phi \partial_j \phi \right]. \quad (156)$$

Define the gravitational and matter momentum densities and the lapse combination:

$$D_i^g = -2 h_{ij} \nabla_k \pi^{kj}, D_i^m = \Pi \partial_i \phi, \quad (157)$$

$$\omega_i = N_1 \partial_i N_2 - N_2 \partial_i N_1, \omega^i = h^{ij} \omega_j. \quad (158)$$

The three dimensional evolution bracket

Use compact variations or periodic fields so that boundary terms vanish. The kinetic momentum derivative is

$$\frac{\delta H_g[N]}{\delta \pi^{ij}} = \frac{2aN}{\sqrt{\det h}} (\pi_{ij} - \lambda_H \pi h_{ij}). \quad (159)$$

Write R_3^{ij} for the spatial Ricci tensor with both indices raised by h^{ij} . The curvature variation contains the lapse derivatives that survive antisymmetrization:

$$\frac{\delta H_{g,\text{spatial}}[N]}{\delta h_{ij}} = b\sqrt{\det h} \left[N \left(R_3^{ij} - \frac{1}{2} h^{ij} R_3 \right) - \nabla^i \nabla^j N + h^{ij} \nabla^2 N \right]. \quad (160)$$

Terms proportional to $N_1 N_2$ cancel in the bracket. Combining the displayed derivatives leaves, before integrating by parts,

$$2ab \int d^3x \left[-\pi^{ij} + (1 - 2\lambda_H) \pi h^{ij} \right] (N_2 \nabla_i \nabla_j N_1 - N_1 \nabla_i \nabla_j N_2). \quad (161)$$

The bracketed lapse Hessians equal $-\nabla_{(i} \omega_{j)}$. Integration by parts gives a momentum-generator term and a trace-gradient remainder:

$$\{H_g[N_1], H_g[N_2]\} = ab D_g[\omega^i] + 2ab(1 - 2\lambda_H) \int d^3x \omega^i \nabla_i \pi. \quad (162)$$

The scalar matter bracket gives $D_m[\omega^i]$. Mixed matter-gravity terms cancel because this matter Hamiltonian has no spatial derivatives of the metric: its metric variation and the gravitational momentum variation have only the symmetric product $N_1 N_2$. Therefore

$$\{H[N_1], H[N_2]\} = ab D_g[\omega^i] + D_m[\omega^i] + 2ab(1 - 2\lambda_H) \int d^3x \omega^i \nabla_i \pi. \quad (163)$$

For arbitrary canonical configurations, common geometric evolution requires the same spatial displacement for field and matter and no extra trace-gradient generator. Thus

$$ab = 1, \lambda_H = \frac{1}{2}. \quad (164)$$

The Hamiltonian trace coefficient is $1/2$, whereas the trace-square coefficient in the Lagrangian is one. A Legendre transform relates them. Reciprocal rescaling $a = 1/\kappa$, $b = \kappa$ preserves the bracket, consistently leaving one overall coupling. The scalar probe does not represent a new gravitational field degree of freedom.

Verification of the three dimensional bracket

The full functional bracket was evaluated on periodic grids with 12^3 , 20^3 and 28^3 points, with fields varying in all three spatial coordinates. The metrics were spatially varying and conformally flat, while the momenta were general symmetric tensors. Separate finite variations included nonconformal diagonal and off-diagonal metric changes.

a	b	λ_H	Common evolution residual
1.0	1.0	0.50	-1.288e-15
0.5	2.0	0.50	-1.288e-15
1.0	1.0	0.35	+8.041e-02
1.1	1.0	0.50	-1.784e-01

Across the twelve bracket evaluations, the full calculation and the analytic formula agreed to $1.33\text{e-}15$. Six independent fourth-order finite-difference functional variations agreed with the analytic derivatives to $4.05\text{e-}11$. These are residuals in dimensionless test configurations, not measurement uncertainties.

The retained coefficients and their reciprocal overall rescaling satisfy common evolution. Changing the trace coefficient or the relative matter-field evolution coefficient gives a nonzero residual that persists with grid refinement. These tests check the implementation of the analytic bracket, rather than sample every possible spatial metric.

The calculation establishes one minimal local completion of the spherical response. Its kinetic coefficient comparison selects $F = \kappa R$ before exterior screen matching, and its spatial term and evolution bracket are mutually consistent. The resulting static equations yield the constant exterior charge, after which the retained source normalization supplies its physical interpretation.

The scope of this result is the field content, derivative class, physical metric and common evolution requirement stated in the derivation. The source law supplies the normalization of the source; the local completion supplies the radial response. Neither the scalar invariant alone nor independence from an arbitrary screen has been substituted for those dynamical requirements.

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